CSE 219 — Signals and Linear Systems

Lec 0–2 · DTFT and DFT, DFT Properties and Circular Convolution

Lesson notes · Exam approach · TF practice with solutions · Quick notes (Ashraf Sir's portion)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

Contents

Exam Map — What the TF Papers Ask — DFT questions and how much they matter
Lesson 1 — From Fourier Series to the DFS and DFT — sampling a periodic signal, N unique coefficients, the DFT pair
Lesson 2 — The DTFT, and the DFT as Its Samples — 2π-periodic spectrum, inverse DTFT, zero-padding, bins → Hz
Lesson 3 — DFT Properties — the clock (mod N), circular shift, modulation, symmetry, Parseval
Lesson 4 — Circular Convolution and Filtering with the DFT — multiply DFTs = circular convolution; zero-pad to L + M − 1
Quick Notes — DTFT and DFT — merged revision notes + key diagrams

> **NOTE** Slide numbers are PDF page numbers of Lecture 1 - DTFT and DFT.pdf (38 pages, “Lec 1”) and Lecture 2 - DFT Properties and Applications.pdf (26 pages, “Lec 2”); the printed slide number on each page is one less. Lecture 0 (Overview of the Transforms, 41 pages) is history and says “Nothing here is examinable”; its one useful picture — the four-transform square — is in Lesson 1.

Exam Map — What the TF Papers Ask

Session (paper)	Questions answered by this note
2023-24 (CSE 219)	Q5(b) Linear convolution of {1,2} and {1,1} using the DFT (10)
2016-17 (CSE 311)	Q3(a) Derive the DFT from G(f); $f_0 = 1/T_0$; compute a DFT; choosing T_s and T_0 (20)

EXAM FOCUS Only two TF questions come straight from these decks (CSE 219 2023-24 Q5(b): linear convolution through the DFT, 10 marks; CSE 311 2016-17 Q3(a): derive and compute a DFT, 20 marks). But every FFT, Bluestein and sampling question (Lessons 5–12) is built on the DFT pair, W_N, the bin reading and circular convolution. Learn: the DFT/IDFT pair by heart, the 4-point W_4 table, X[N − k] = X*[k], and the zero-padding rule N ≥ L + M − 1.

Lesson 1 — From Fourier Series to the DFS and DFT

Lec 0 sl. 22; Lec 1 sl. 2–19

1. Four Fourier Transforms, One Rule (Lec 0 sl. 22; Lec 1 sl. 2)

	continuous time	discrete time
periodic	CTFS: discrete, aperiodic spectrum a_k	DFS/DFT: discrete, periodic spectrum X[k]
aperiodic	CTFT: continuous, aperiodic $X(j\omega)$	DTFT: continuous, periodic $X(e^{j\omega})$

MEMORIZE Discrete in one domain ⇔ periodic in the other. Sampling time makes the spectrum periodic; keeping only N frequency bins makes time N-periodic. The DFT is the only one a computer can compute.

2. Sampling a Periodic Signal (sl. 5–10)

x(t) has period T_0; take N samples per period, $T_0 = NT_s$, $x[n] = x(nT_s)$ (N-periodic). Represent the sampling as $x_p(t) = x(t)p(t) = \Sigma\, x[n]\delta(t - nT_s)$. Its CTFS coefficients: the impulses sift the integral into a finite sum.

```
c_k = (1/T₀)∫_{T₀} x_p(t)e^{-jkω₀t}dt = (1/T₀)Σ_{n=0}^{N-1} x[n]e^{-jkω₀nT_s},   ω₀T_s = 2π/N
    = (1/T₀)Σ_{n=0}^{N-1} x[n]e^{-j2πkn/N}
c_{k+N} = c_k·e^{-j2πn} = c_k   ⇒ only N distinct coefficients (sampling in time made the
spectrum periodic)
c_{-k} = c_{N-k}  ⇒ negative frequencies fold onto the upper bins
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3. The Inverse by Orthogonality (sl. 12–14)

```
Σ_{k=0}^{N-1} e^{j2πk(n-m)/N} = N if n = m,  0 if n ≠ m   (geometric series: (1 -
e^{j2π(n-m)})/(1 - e^{j2π(n-m)/N}) = 0)
⇒ x[n] = (1/N)Σ_k (T₀c_k)e^{j2πkn/N}
```

4. DFS and DFT (sl. 15–19)

$$\text{DFS: } a_k = (1/N)\sum_{n=0}^{N-1} \tilde{x}[n]e^{-j2\pi kn/N},\quad \tilde{x}[n] = \sum_{k=0}^{N-1} a_k\, e^{j2\pi kn/N}$$

$$\text{DFT: } X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn},\quad x[n] = (1/N)\sum_{k=0}^{N-1} X[k]W_N^{-kn},\quad W_N = e^{-j2\pi/N}$$

Bridges: $X[k] = N\cdot a_k$ (DFT = DFS rescaled); $c_k = X[k]/T_0$ (CTFS of the sampled periodic signal). The DFT never needs the continuous signal: any list of N numbers has one (sl. 18). Reading the bins: k = 0 … N/2 are positive frequencies (low → high); k = N/2 … N − 1 are the folded negative frequencies (high → low).

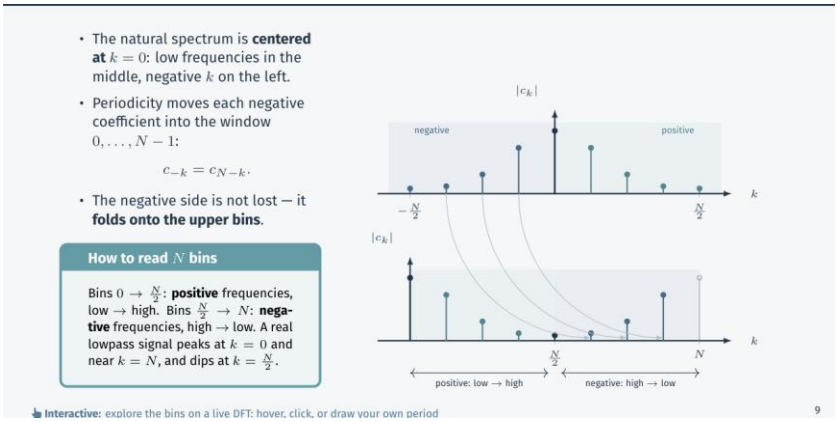


Figure 1.1 — Negative-frequency coefficients fold onto the upper bins (Lec 1, sl. 11)

Exam Approach

1. Derivation: sample, write x_p as an impulse train, sift the CTFS integral, use $\omega_0 T_s = 2\pi/N$.
2. Prove N-periodicity of the coefficients ($k \to k + N$).
3. Inverse: multiply by $e^{j2\pi kn/N}$, sum over k, use the orthogonality sum.
4. State the bin ↔ frequency map.

Practice — Lesson 1

(a) Actual — CSE 311 2016-17 Q3(a). (b) Constructed (Lec 1, last slide).

(a) g(t) is sampled over $0 \le t \le T_0$ at intervals T_s giving N_0 samples; $G_q = \Sigma_{k=0}^{N_0-1} g_k e^{-jq\Omega_0 k}$, $g_k = T_s\, g(kT_s)$, $G_q = G(qf_0)$, $\Omega_0 = 2\pi f_0 T_s$. (i) Derive this from G(f). (ii) Prove $f_0 = 1/T_0$. (iii) Compute the DFT for g(0..3) = 1, 2, 3, 4 (t = 0, 1, 2, 3); give the sample frequencies and G(f). (iv) How should T_s and T_0 be chosen? (4 + 5 + 5 + 6)

(b) Compute the 4-point DFT of {1, 0, −1, 0}. Which bins light up and why?

✓ Solution

```
(i) G(f) = ∫₀^{T₀} g(t)e^{-j2πft}dt ≈ Σ_{k=0}^{N₀-1} g(kT_s)e^{-j2πfkT_s}·T_s   (rectangle
rule, time-limited g)
    evaluate at f = qf₀:  G(qf₀) = Σ g_k e^{-j2πqf₀kT_s} = Σ g_k e^{-jqΩ₀k},  Ω₀ = 2πf₀T_s
(ii) Sampling in time makes G_q periodic; requiring the N₀ frequency samples to cover exactly
one period
     (G_{q+N₀} = G_q) needs e^{-jN₀Ω₀k} = 1 for all k ⇒ N₀Ω₀ = 2π ⇒ 2πf₀T_sN₀ = 2π ⇒ f₀ =
1/(N₀T_s) = 1/T₀
(iii) T_s = 1, N₀ = 4 ⇒ T₀ = 4 s, f₀ = 0.25 Hz, Ω₀ = π/2, W = e^{-jπ/2} = -j
     G₀ = 1+2+3+4 = 10          f = 0
     G₁ = 1 − 2j − 3 + 4j = −2+2j   f = 0.25 Hz
     G₂ = 1 − 2 + 3 − 4 = −2       f = 0.5 Hz (= f_s/2)
     G₃ = 1 + 2j − 3 − 4j = −2−2j   f = 0.75 Hz ≡ −0.25 Hz  (G₃ = G₁*)
(iv) T_s ≤ 1/(2B) to avoid aliasing (B = essential bandwidth);  T₀ long enough to capture g
(time-limited) and
     to give the needed resolution f₀ = 1/T₀;  N₀ = T₀/T_s, preferably a power of 2 for the FFT
(zero-pad g)
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(b) X = {0, 2, 0, 2}: x = cos(πn/2) has frequency ω = π/2 = bin 1 and its negative partner −π/2 ≡ bin 3 (cos = ½$e^{jπn/2}$ + ½$e^{-jπn/2}$, each with weight N/2 = 2).

Lesson 2 — The DTFT, and the DFT as Its Samples

Lec 1 sl. 20–34

Aperiodic finite sequence x[n] (0 ≤ n ≤ N − 1). CTFT of the impulse train $x_p(t) = \Sigma x[n]\delta(t - nT_s)$: $X_p(j\omega) = \Sigma\, x[n]e^{-j\omega nT_s}$. Relabel $\omega T_s \to \omega$ (digital frequency, rad/sample):

$$\mathbf{X(e^{j\omega}) = \Sigma_n\, x[n]e^{-j\omega n} \qquad x[n] = (1/2\pi)\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega}$$

- 2π-periodic: $X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$ since $e^{-j2\pi n} = 1$ (sl. 25). Principal interval $[-\pi, \pi] \leftrightarrow$ physical $[-\omega_s/2, \omega_s/2]$.
- Inverse (sl. 26): substitute the analysis sum, swap sum and integral, use $(1/2\pi)\int_{-\pi}^{\pi} e^{j\omega r}d\omega = \delta[r]$.
- DFT = DTFT samples (sl. 29): $X[k] = X(e^{j\omega})|_{\omega=2\pi k/N}$. Periodic view: FS coefficients; aperiodic view: N samples of the DTFT — the same N numbers.
- Zero-padding to $M \geq N$ (sl. 31): same DTFT, sampled at spacing $2\pi/M$ — more pixels, not more information.
- Bins to Hz (sl. 33–34): $f_k = k\, f_s/N$ for $0 \leq k \leq N/2$, $(k - N)f_s/N$ above; $\Delta f = f_s/N = 1/T_0$ (numpy.fft.fftfreq order).

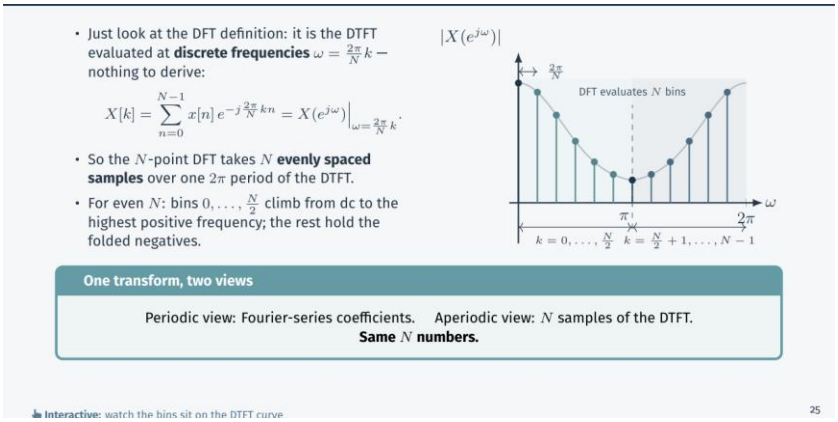


Figure 2.1 — The N-point DFT samples one period of the DTFT (Lec 1, sl. 29)

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Worked example (sl. 32): x = {1, 2, 3, 4}, N = 4, W₄ = −j
W₄^{kn}:   k=0: 1  1  1  1 |  k=1: 1 −j −1  j |  k=2: 1 −1  1 −1 |  k=3: 1  j −1 −j
X = {10, −2+2j, −2, −2−2j};  |X| = {10, 2√2, 2, 2√2};  X[3] = X*[1] (bin 3 = bin 1's negative partner)
N = 8, f_s = 8 kHz: Δf = 1 kHz;  bins 0..4 → 0..4 kHz,  bins 5,6,7 → −3, −2, −1 kHz
```

Exam Approach

1. Write W_N and its powers ($W_4 = -j$, $W_8 = (1 - j)/\sqrt{2}$).
2. Tabulate W_N^{kn} (it is periodic in kn mod N).
3. Sum row by row; check $X[0] = \Sigma x$ and $X[N - k] = X^*[k]$ for real x.
4. Convert bins to Hz with f_s/N.

Lesson 3 — DFT Properties

Lec 2 sl. 2–12

Everything lives on a clock: indices are taken mod N, $((n))_N = n \bmod N$, e.g. $((9))_8 = 1$, $((-1))_8 = 7$; the DFT sees the periodic extension $\tilde{x}[n] = x[((n))_N]$. One proof move gives every property: substitute into the DFT sum, re-index, and let the clock absorb whatever spills past the edge.

Property	Time	DFT
linearity	$a\,x[n] + b\,y[n]$ (same N)	$a\,X[k] + b\,Y[k]$
circular time shift	$x[((n-m))_N]$	$W_N^{km}\,X[k]$ (phase only)
modulation	$W_N^{-k_0 n}\,x[n]$	$X[((k-k_0))_N]$; $(-1)^n x \leftrightarrow X[k - N/2]$ (low ↔ high)
time reversal	$x[((-n))_N]$	$X[((-k))_N]$
conjugation	$x^*[n]$	$X^*[((-k))_N]$
real signal	x real	$X[N-k] = X^*[k]$: $\lvert X\rvert$ even, $\angle X$ odd about N/2
duality	$X[n]$	$N\,x[((-k))_N]$
circular convolution	$x \circledast h$	$X[k]H[k]$
multiplication	$x[n]h[n]$	$(1/N)\,X \circledast H$
Parseval	$\Sigma\lvert x[n]\rvert^2$	$(1/N)\Sigma\lvert X[k]\rvert^2$

```
Circular shift proof: Σ x̃[n − m]W^{kn} = W^{km} Σ_{r=−m}^{N−1−m} x̃[r]W^{kr} = W^{km}X[k] (any N consecutive terms)
Parseval check on {1,2,3,4}: 1+4+9+16 = 30 = (100 + 8 + 4 + 8)/4 ✓;  |X[k]|²/N = energy in bin k
```

TRAP • Shifts are circular: $x[n-m]$ on N points wraps the tail to the front. • Linearity needs equal lengths — zero-pad the shorter sequence first. • Real and even $\tilde{x} \Rightarrow X$ real.

Lesson 4 — Circular Convolution and Filtering with the DFT

Lec 2 sl. 13–23

Linear convolution y = x * h has length L + M − 1. Multiply two N-point DFTs and invert:

```
y[n] = (1/N)Σ_k X[k]H[k]W^{-kn} = Σ_m x[m]·[(1/N)Σ_k H[k]W^{-k(n-m)}] = Σ_{m=0}^{N-1} x[m]
h[((n - m))_N]
```

$$(x \circledast h)[n] = \sum_{m=0}^{N-1} x[m]\, h[((n-m))_N] \;\Leftrightarrow\; X[k]H[k]$$

Two-wheel picture (sl. 16): x on the outer wheel, h reversed on the inner wheel; rotate one tick per output, multiply across the spokes, add. The off-edge taps of linear convolution (which multiply nothing) are instead wrapped onto x's tail — that wrap (time aliasing) is the only difference.

```
Example (sl. 21): x = {2,4,5,6,5,3,2,1}, h = {1,2,1}
linear (length 10): {2, 8, 15, 20, 22, 19, 13, 8, 4, 1}
circular N = 8:    {6, 9, 15, 20, 22, 19, 13, 8}   — y[0] = 2+4, y[1] = 8+1: the tail {4, 1}
wrapped onto the front
```

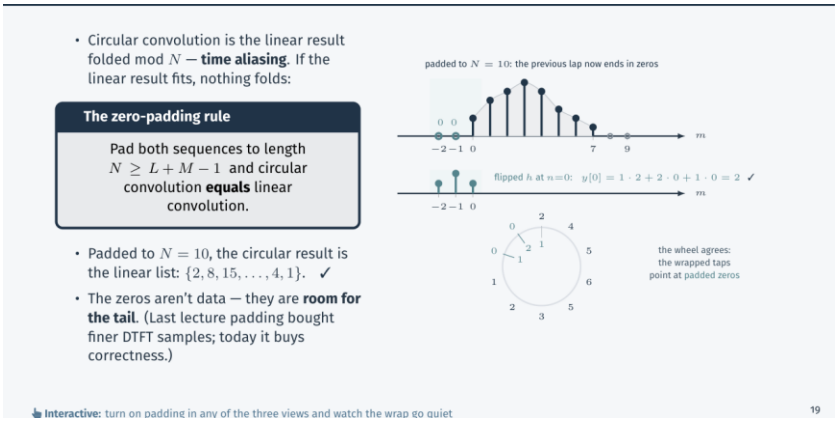


Figure 4.1 — Zero-padding to N = 10 parks the wrapped taps on zeros (Lec 2, sl. 22)

$$\text{Circular} = \text{linear} \;\Leftrightarrow\; N \geq L + M - 1 \;\; \text{(zero-pad both)}$$

```
Filtering recipe (sl. 23): 1) N ≥ L + M − 1, pad x and h   2) X = DFT{x}, H = DFT{h}
                           3) Y = X·H bin by bin              4) y = IDFT{Y}; keep L + M − 1
samples
Cost with direct DFTs ~N² each (loses to L·M); with the FFT ~N log₂N (Lesson 8) — the fastest
known way
```

Exam Approach

1. Pick N ≥ L + M − 1 (a power of 2 if an FFT is asked).
2. Zero-pad, compute both DFTs (W_4 table).
3. Multiply bin by bin.
4. IDFT: x[n] = (1/N)Σ X[k]W^{-kn}; check against the direct convolution.

Practice — Lesson 4

(a) Actual — CSE 219 2023-24 Q5(b). (b) Practice set 4.1, 4.4.

(a) x = {1, 2}, h = {1, 1}. Show step by step the linear convolution using the naive DFT method. (10)

(b) {1, 2} * {1, −1} with a 4-point DFT; {1, 0, 1} * {1, −1, 2} with a 4-point DFT (show the wrap).

✓ Solution

```
(a) L + M − 1 = 3 ⇒ use N = 4 (N = 3 also works). Pad: x = {1,2,0,0}, h = {1,1,0,0}; W₄ = −j
    X[k] = Σ x[n](−j)^{kn}:  X = {3, 1 − 2j, −1, 1 + 2j}
    H[k]:                    H = {2, 1 − j, 0, 1 + j}
    Y = XH = {6, (1−2j)(1−j) = −1 − 3j, 0, −1 + 3j}
    y[n] = ¼Σ Y[k] j^{kn}:
    y[0] = ¼(6 − 1 − 3j + 0 − 1 + 3j) = 1
    y[1] = ¼(6 + (−1−3j)j + 0 + (−1+3j)(−j)) = ¼(6 + 3 − j + 3 + j) = 3
    y[2] = ¼(6 − (−1−3j) + 0 − (−1+3j)) = ¼(6 + 1 + 3j + 1 − 3j) = 2
    y[3] = ¼(6 + (−1−3j)(−j) + 0 + (−1+3j)(j)) = ¼(6 − 3 + j − 3 − j) = 0
    y = {1, 3, 2} (+ a trailing 0) = direct x * h ✓
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```
(b) {1,2}*{1,−1}: X = {3, 1−2j, −1, 1+2j}, H = {0, 1+j, 2, 1−j} ⇒ y = {1, 1, −2, 0}
    {1,0,1}*{1,−1,2}: linear = {1, −1, 3, −1, 2} (N ≥ 5 ⇒ use 8); with N = 4 the last sample
wraps:
    circular = {1 + 2, −1, 3, −1} = {3, −1, 3, −1}
```

Quick Notes — DTFT and DFT

Revision points for Lessons 1–4.

L1 · DFS/DFT

- $c_k = (1/T_0)\Sigma x[n]e^{-j2\pi kn/N}$; $c_{k+N} = c_k$; $c_{-k} = c_{N-k}$.
- $X[k] = \Sigma x[n]W^{kn}$, $x[n] = (1/N)\Sigma X[k]W^{-kn}$, $W_N = e^{-j2\pi/N}$; $X[k] = N\, a_k$.

L2 · DTFT

- $X(e^{j\omega}) = \Sigma x[n]e^{-j\omega n}$, 2π-periodic; inverse $(1/2\pi)\int_{2\pi}$.
- $X[k] = X(e^{j2\pi k/N})$; zero-padding = finer samples; $f_k = k\, f_s/N$.

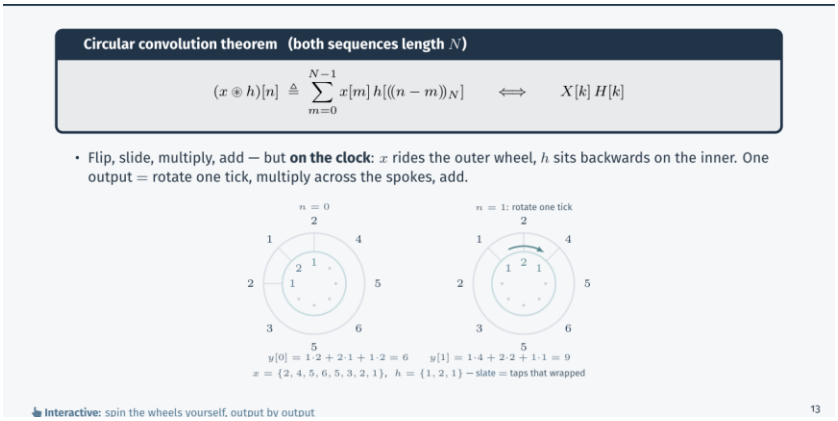
L3 · Properties

- circular shift $\leftrightarrow W^{km}X[k]$; real $x \Rightarrow X[N-k] = X^*[k]$; Parseval $\Sigma|x|^2 = (1/N)\Sigma|X|^2$.

L4 · Circular conv

- $XH \leftrightarrow x \circledast h$; circular = linear iff $N \geq L + M - 1$.
- $\{1,2\} * \{1,1\} = \{1,3,2\}$ via 4-point DFTs: $Y = \{6, -1-3j, 0, -1+3j\}$.

Key Diagrams to Reproduce



Circular convolution: two wheels (Lec 2, sl. 16)